

The UE is connected to more than one base station and transmitting/receiving is both at the same time. The CoMP algorithm resolves duplicate packets received. At the cell edge, when moving from one cell to another, the handover time is very low, and the packet loss is minimised. This improves robustness of the transmission, and the connection is more reliable. This results in no dropped video frames in the transmission from the cars.

**Interference.** The cars typically have several sensors that take measurements like acceleration, speed, temperature, vibration, and tire pressure. These sensors can create electromagnetic interference, which can hinder the reception and transmission of the 5G radio signals.

**Solution.** The antennas need to be mounted strategically on the racing cars so that they are not prone to the interference from other components in the car. The radio frequency circuit needs to be designed with a high level of tolerance against spurious signals.

**Bandwidth.** The racing cars are travelling at a very high speed, hence

a lower modulation and coding scheme (MCS) is required to be used to provide a robust transmission and reception of signals. Similarly, a robust code rate with more redundant bits is needed for the forward error correction (FEC) to recover from any transmission errors.

Hence the bandwidth required can be as large as 200MHz to provide the required throughput for the racing cars. This raises the issue of which spectrum to use where a sufficiently large bandwidth is available

**Solution.** The unlicensed 5GHz or 6GHz bands are an option where a large bandwidth can potentially be available for use. However, this band is currently also shared for Wi-Fi, and hence the reliability and guaranteed availability of this spectrum may be an issue.

Another option is to use the custom band like the 10GHz band, which is currently used by the military for radar stations. The requirement is to keep the transmit power low so that there is no interference to incumbent users in this band.

Custom radios and trackside infrastructure would need to be developed when using these bands to provide the required quality of service and reliability for the racing cars.

## 5G spectrum

One of the key decisions for the solution is the choice of spectrum to be used for the 5G system, so that the system can operate in all the countries while providing consistent

KPIs like throughput and latency. So, following are the options for the operating frequency of the solution, along with the pros and cons of each option.

**3.5GHz mid band.** The 3.5GHz mid-band spectrum is crowded and is allocated to the operators in most countries, with the operators having about 20 to 50MHz bandwidth. There are a few countries, such as Spain, where the individual operators like Telefonica, Vodafone, and Orange have 90 to 100MHz of bandwidth. In Great Britain the operators have about 20 to 60MHz of bandwidth each. In countries like Bahrain, Canada, Russia, Netherlands, and Turkey the 3.5GHz spectrum has not yet been auctioned.

Hence to operate in the 3.5GHz band, we would need to have partnerships with two or more operators to be able to get the 100MHz spectrum needed. Some countries have reserved part of spectrum in the mid band for unlicensed and shared use (not available universally), as shown in Table 1. The available spectrum is primarily for private indoor use and should not interfere with incumbents operating in the same frequency band.

**mmWave bands.** The licensed mmWave spectrum is available in a few different bands: 26GHz, 28 GHz, 37GHz, 39GHz, and 47GHz. Typically, the amount of frequency allocated to each operator in the mmWave bands is higher—from 200MHz to 400MHz or more. This is ideal in terms of a large chunk of bandwidth required for the motor-sport solution. Some of the unlicensed bands are:

- USA: 37 to 37.6GHz (3x200MHz) has been allocated by FCC for shared/unlicensed use

- Germany, UK, Australia: 24.25 to 27.5GHz for local licenses

- Japan: 28.3 to 29.1GHz (150MHz outdoor use) for local license **EFY**

**TABLE 1**  
**PRIVATE USE SPECTRUM**

| Country | Band (MHz)       |
|---------|------------------|
| Germany | 3700-3800 (100M) |
| USA     | 3550-3700 (150M) |
| UK      | 3800-4200 (400M) |
| Canada  | 3500 planned     |
| Japan   | 4500-4800 (300M) |

**TABLE 2**  
**5G NR FR2 FREQUENCY BANDS**

| Band | Frequency (GHz) | Common Name | Uplink/Downlink (TDD) | Channel Bandwidth (MHz) |
|------|-----------------|-------------|-----------------------|-------------------------|
| n257 | 28              | LMDS        | 26.50 – 29.50         | 50, 100, 200, 400       |
| n258 | 26              | K-band      | 24.25 – 27.50         | 50, 100, 200, 400       |
| n259 | 41              | V-band      | 39.50 – 43.50         | 50, 100, 200, 400       |
| n260 | 39              | Ka-band     | 37.00 – 40.00         | 50, 100, 200, 400       |
| n261 | 28              | Ka-band     | 27.50 – 28.35         | 50, 100, 200, 400       |